Course Code	Course Title	L	T	P	C
BAMAT207	Probability and Statisics	3	0	2	4
Pre-requisite	BAMAT101	**Syllabus Version**			
		1			

Course Objectives

Apply probability and statistical methods to solve engineering problems, including system reliability, risk assessment, and quality control.
To develop predictive models and analyze data using probability distributions, correlation, regression, and hypothesis testing for engineering applications.
To design experiments and interpret variability using ANOVA and structured experimental designs to optimize engineering systems and innovations.

Course Outcomes

Apply probability concepts, including axioms, conditional probability, and Bayes' theorem, to solve real-world problems.
Implement probability distributions to model random variables and analyze statistical properties.
Utilize hypothesis testing methods to evaluate statistical significance in experiments.
Demonstrate correlation and regression techniques for data analysis and predictive modeling.
Analyze ANOVA methods and experimental designs to interpret variability in datasets.

Module:1	**Probability and Random Variables**	**9 hours**

Basic Probability- Axioms, probability spaces, conditional probability, Bayes' theorem. Random Variables and Distributions - Discrete and Continuous random variables, probability mass functions, probability density functions. Joint Distributions - Joint, marginal, and conditional distributions. Expectation and Variance – Moments, variance, covariance, correlation and independence. Functions of one-dimensional random variables.

Module:2	**Probability Distributions**	**8 hours**

Bernoulli, Binomial, Poisson, Geometric, Negative Binomial Distribution, Uniform, Normal, Exponential, Gamma, Weibull distributions. Central limit theorem, Sampling Distributions – t, F, Chi-square (Concept Only).

Module:3	**Testing of Hypothesis**	**9 hours**

Formulating and testing hypotheses, Simple and composite hypotheses, Type I and Type II errors. Parametric Tests – Z, t and F-test of hypothesis. Chi-square test for

Goodness of fit and independence of attributes. Confidence Intervals- Constructing and interpreting confidence intervals.		
Module:4	**Correlation and Regression**	**9 hours**
Pearson's correlation and Spearman's rank correlation coefficient. Multiple and partial correlation. Simple and Multiple linear regression: model, assumptions, estimation and interpretation of regression coefficients. Model diagnostics and validation.		
Module:5	**ANOVA and Design of Experiments**	**8 hours**
One-way and Two-way Analysis of Variance – CRD, RBD and LSD of experiments.		
Module:6	**Contemporary Topics**	**2 hours**
	Total Lecture Hours:	**45 hours**

Text Book(s)

Sheldon M. Ross, "**Introduction to Probability and Statistics for Engineers and Scientists**", Academic Press, 6th Edition, **2021**
Richard Arnold Johnson, Irwin Miller, John E. Freund, "**Probability and Statistics for Engineers**", Pearson Education, 9th Edition, **2018**

Reference Books

V. K. Rohatgi and A. K. Md. Ehsanes Saleh, "**An Introduction to Probability and Statistics**", John Wiley and Sons, 3rd Edition, **2015**
Douglas C. Montgomery and George C. Runger, "**Applied Statistics and Probability for Engineers**", John Wiley and Sons, 7th Edition, **2018**

Indicative Experiments

1. Introduction, understanding data types, importing and exporting data sets.	2 hours
2. Exploratory Data Analysis - Tabulation and Data Visualization - Create histograms, box plots, and scatter plots to visualize data distributions and relationships.	4 hours
3. Exploratory Data Analysis - Simulation and Descriptive Statistics - Summarize data using measures of central tendency, variability, skewness and kurtosis.	4 hours
4. Fitting of Probability Distributions – Discrete and Continuous	2 hours
5. Testing of Hypothesis and Confidence Intervals – Z- test and t - test.	2 hours

6. Testing of Hypothesis and Confidence Intervals – F and Chi-square test of hypothesis	2 hours
7. Simple linear Regression and Correlation Analysis – Modelling, estimating and interpreting correlation and regression coefficients.	4 hours
8. Multiple Linear Regression and Correlation Analysis – Modelling, estimating and interpreting correlation and regression coefficients.	4 hours
9. Performing One-way ANOVA – CRD - Modelling, estimation and interpretation of Coefficients.	2 hours
10. Performing Two-way ANOVA – RBD and LSD- Modelling, estimation and interpretation of Coefficients.	4 hours
Total Laboratory Hours:	**30 hours**

Text Book(s)

Maria Dolores Ugarte, Ana F. Militino, and Alan T. Arnholt, "**Probability and Statistics with R**", CRC Press, 2nd Edition, **2016**

Reference Books

Tilman M. Davies, "**The Book of R: A First Course in Programming and Statistics**", No Starch Press, 1st Edition, **2016**

Dieter Rasch, Rob Verdooren, and Jürgen Pilz, "**Applied Statistics: Theory and Problem Solutions with R**", Wiley, 1st Edition, **2019**

Mode of Evaluation : Continuous Assessment Test, Digital Assignment, Quiz, Final Assessment Test, Lab Continuous Assessment, Lab Final Assessment

Recommended by Board of Studies :	14-05-2025
Approved by Academic Council : No. 78	12-06-2025